

Nathan Bakare

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EDUCATION

University of Houston

Houston, TX

Bachelor of Science in Computer Engineering | GPA: 3.87/4.0

Expected May 2028

- Relevant Coursework: Circuit Analysis, Data Structures & Algorithms, Fundamentals of Software Engineering, Operating Systems, Database Systems, Object Oriented Programming

TECHNICAL SKILLS

Languages: Python, Java, C, HTML/CSS, JavaScript, TypeScript, SQL

Frameworks/Libraries: React, Next.js, Node.js, Express.js, FastAPI, Flask, PyTorch, XGBoost, Tailwind CSS

Tools/Platforms: Git, GitHub, Docker, MLflow, Airflow, GraphQL API, Langflow, OpenSearch, Docling, pytest, Vitest

EXPERIENCE

IBM

San Jose, CA

Software Developer Co-op

Aug. 2026 – Present

- Resolved user-facing bugs across OpenRAG's **TypeScript/Next.js** frontend, fixing a stale-state race condition that caused search bar buttons to require two clicks, surfacing inline error tooltips on failed Knowledge table rows, and correcting a **WCAG 1.4.11** contrast failure in the data grid.
- Diagnosed and resolved backend reliability bugs in OpenRAG's **Python/FastAPI** ingestion pipeline, closing a filename-resolution gap across 6 document connectors (Google Drive, SharePoint, OneDrive, S3, Azure Blob, IBM COS) and adding retry logic to **OpenSearch** bulk indexing that eliminated false failure reports on files that had actually indexed successfully.
- Designed a proposed event-driven ingestion architecture connecting **Confluent Kafka** to OpenRAG's ingestion gateway and **IBM WatsonX** embedding models, replacing batch processing with real-time CREATE/UPDATE/DELETE event handling and a content-aware router across three embedding models.

JP Morgan Chase

Plano, TX

Software Engineer Intern

Jun. 2026 – Aug. 2026

- Built a full-stack AI cost intelligence dashboard (**React/TypeScript**) processing **3.1M+** rows of LLM spend data through an offline **Node.js** ETL pipeline, supporting real-time multi-dimensional filtering without shipping raw data to the browser.
- Wrote a unit-tested (**Vitest**) analytics engine that computes spend anomaly detection, budget forecasting, and ranked cost-saving opportunities, used by finance stakeholders to review **FinOps** spend.
- Integrated **Databricks Genie** through a secure **Vite** server-side proxy and built client-side export to Excel/PDF/PowerPoint, cutting manual reporting steps for the cost-governance team.

PROJECTS

CourseMate | HTML, CSS, JavaScript, GraphQL API

Jan. 2026

- Built a Chrome extension that surfaces RateMyProfessors ratings directly inside UH's course registration flow, removing the need to tab-switch between the two sites.
- Developed an async web scraping system to fetch and cache ratings for **2,000+** UH professors.
- Implemented local caching + rate limiting (1 req/sec) to cut external lookups by **30%**, reducing latency and staying within provider rate limits.

The Move – RSVP Component | React, TypeScript, Tailwind CSS, Node.js, JWT Auth

May 2025

- Designed and built an inline RSVP feature (Yes/No/Maybe) for a social event-sharing application.
- Developed real-time RSVP updates, load-tested to **10+** simultaneous users.

E-Commerce Demand Forecasting System | Python, PyTorch, Flask, Docker, XGBoost

Mar. 2025

- Trained an **LSTM** model in PyTorch on **500K+** sales records to forecast 30-day demand across 15 categories of products selling 5,000 units/day, achieving an MAE of 2 units.
- Built an **MLOps** pipeline (Airflow + MLflow) automating feature extraction from sales data and customer-review text (BERT NLP) for model retraining.
- Deployed an **XGBoost** classifier via Flask API and Docker to flag stockout risk for inventory restocking, evaluated with **89% precision** on held-out data.